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AutoDock Tutorial

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Authors: Ana C. Murrieta and Flavio F. Contreras-Torres. Tecnológico de Monterrey.

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version v1.0

INTRO

This notebook is a **hands-on introduction** to molecular docking using AutoDock and to the essential concepts required to perform a basic docking workflow. It is intended for beginners as well as for learners with prior experience who wish to strengthen their understanding of ligand–receptor docking, structure preparation, docking setup, and interpretation of computational results.

The notebook is divided into three main parts:

- **Part 1: Introduction to Molecular Docking.** You will become familiar with the basic principles of molecular docking, the general purpose of AutoDock, and the role of docking in the computational analysis of ligand–receptor interactions.
- **Part 2: Protocol.** You will learn the main steps of a typical docking workflow, including the selection of receptor and ligand structures, preparation of input files, definition of docking parameters, execution of docking calculations, and initial analysis of the results.
- **Part 3: Results Interpretation and Practice.** You will learn how to examine predicted binding poses, evaluate docking outputs, identify common issues, and reinforce the workflow through guided practice.

Sources and Learning Materials

This tutorial is not meant to be the first or the last resource you will use to learn **AutoDock Vina** and molecular docking. Instead, it is a curated learning path built from several tutorials, lecture notes, exercises, and official documentation.

Many of the ideas, examples, and exercises presented here are inspired by or adapted from existing educational materials, including the official documentation and common teaching resources used in courses and online tutorials. For this reason, you may notice that some exercises or examples look similar to ones you have seen elsewhere. This is intentional: these problems are standard, well-tested ways of learning core concepts.

The goal of this tutorial is not to present completely new material, but to organize and connect these concepts in a coherent, progressive way, with explanations, practice exercises, testing, and debugging techniques all in one place.

You are encouraged to complement this tutorial with other resources, such as:

- The official AutoDock Vina documentation
- Free course notes, books, and lecture materials Vina manual, AutoDock Vina repository, etc.
- Online tutorials jRicciL

Learning works best when you see the **same ideas explained in multiple ways and from multiple sources**.

PART 1. Introduction to Molecular Docking

Molecular docking is a core method in structure-based drug design (SBDD) for predicting, with a substantial degree of accuracy, the binding pose of small molecules in the active site of a receptor [1]. It compromises several computational approaches with the goal to fit ligands into multiple receptor states (e.g., active/inactive, co-activator–bound, mutants), to **predict binding poses** and to **estimate affinity** (i.e., binding affinity). Docking returns a ranked list of compounds and highlights key interactions (e.g., anchor residues, H-bonds, π–π, hydrophobic contacts), supporting *binding-mode hypotheses* and ligand prioritization. Results depend on approximations—software [2] (e.g., AutoDock Vina/Qvina, MOE, Glide, etc.), parameters (e.g., search box, exhaustiveness, number of poses), and the **scoring function** (e.g., the AutoDock Vina empirical score, MOE's London dG scoring function, and others). Robustness improves when using ensembles of receptor structures (i.e., multiple PDB entries, states, or mutants) and verifying that top best-scoring ligands from each docking run can be pooled and de-duplicated.

Molecular docking and **virtual screening** are related but differ in scope and purpose. Docking typically refers to evaluating one (or a small set of) ligand–receptor pairs to predict binding pose(s), estimate affinity, and rationalize key contacts. Virtual screening [3] applies docking (and often other filters) **at scale** to large libraries, ranking compounds to prioritize a small subset for experimental testing. In practice, docking is the *engine* that generates poses and scores; virtual screening is the *workflow* that uses that engine repeatedly—adding library preparation, property/PAINS filters, rescoring or consensus scoring, and enrichment/hit-rate assessment—to identify novel chemotypes.

This is a self-authored tutorial for performing docking calculations with AutoDock Vina. The AutoDock Suite is a free, open–source software for the computational docking and virtual screening of small molecules against macromolecular receptors, developed at the Center for Computational Structural Biology (CCSB), La

Jolla, CA, USA. The suite currently includes complementary tools such as *AutoDock (AD4)*, *AutoDock Vina*, *Raccoon2*, *AutoDockTools*, and *AutoLigand*. The software has been implemented, calibrated, and tested on diverse protein–ligand complexes of biological and medicinal interest. For a comprehensive, step-by-step workflow, see the seminal article [4] at Nature Protocols journal.

Before we continue, a brief note on the differences between **AutoDock (AD4)** and **AutoDock Vina**:

- **Scoring**
  - **AD4:** Empirical free-energy force field (vdW, electrostatics, H-bond, desolvation) + torsional penalty.
  - **Vina:** Compact empirical score function (gaussians + repulsion + hydrophobic + H-bond).
  - *Do not compare scores across engines—only within the same engine/config.*
- **Search**
  - **AD4:** Lamarckian genetic algorithm (LGA) with Solis–Wets local search; tuned via `ga_run`, `ga_pop_size`, `ga_num_evals`, etc.
  - **Vina:** Stochastic global search with rapid, gradient-based local optimization (BFGS-like); tuned via `--exhaustiveness`, `--num_modes`, `--energy_range`, `--seed`.
- **Setup**
  - **AD4:** Requires **AutoGrid** (precomputed affinity maps; AD4 atom types).
  - **Vina:** Builds the grid internally from **PDBQT** (no AutoGrid step).
- **Performance / Use**
  - **AD4:** More configurable; provides per-term energy components.
  - **Vina:** Typically faster and multithreaded—good for large screens.

Note:
Use **AD4** when you need term-level energy decomposition or to reproduce legacy protocols; use **Vina/QVina** for streamlined, high-throughput docking.

In both cases validate the protocol by re-docking ligands into cognate co-crystal structures of **comparable conformational complexity**.

This tutorial focuses on **AutoDock Vina** and related utilities; please ensure they are pre-instaled on your system:

- **Python** - Scripting and process automatization
- **Pymol** - Graphic visualization of molecular structures
- **Chimera** - Molecular visualitazion and docking preparation
- **Open Babel** - Conversion between molecular structure formats
- **Modeller** - Homology modeling of the protein structures
- **AutoDock Vina** — Molecular docking engine
- **Qvina** - Molecular docking engine (AutoDock Vina derivative)
- **CavityPlus** - Identification of binding pockets and cavities
- **LigPlot+** - Visualization of protein-ligand interactions
- **Bash** - Shell scripting for automatization of repetitive tasks

Note:
The instructions adopted in this tutorial are based on a working environment running Ubuntu 22.04.4 LTS.

PART 2. Protocol

The docking methodology —as we apply it— consists of the following steps:

- **Selection of Structures**
- **Preparation of Structures**
- **Docking Setup and Execution**
- **Analysis of Results**

Each step is equally important; however, selecting the appropiate structures and parameters is particularly crucial for obtaining reliable results. In this tutorial, we will go through each step using a simple example of one molecule —rosiglitazone (RGZ)— and one receptor —peroxisome proliferator-activated receptor gamma (PPAR-γ).

2.1 Selection of Structures

Before we event think about docking, we must clearly define the **object** and **subject** of the study — in other words, our **research goal** and the **biological system** under investigation. This step is crucial, as the selection of molecular structures must align with the specific aims of the study.

In this tutorial, for example, our goal is to investigate the agonistic activity on the receptor **PPAR-γ** (the object), given that its activation has been reported to have **therapeutic potential** in regulating metabolic processes involved in adipogenesis, glucose homeostasis, inflammation, and the uptake and storage of fatty acids (the subjects).

We will be working with the **active conformation** of the human PPAR-γ receptor. Therefore, it is essential to select ligands that have been experimentally validated as **agonists** of PPAR-γ.

Such compounds can be identified by consulting curated chemical databases, for instance, ChEMBL, where bioactivity data from experimental assays are publicly available.

Selection of the Receptor

The next step is to select the appropriate molecular structures. In the case of the receptor, the structure should be obtained from **experimentally resolved structural data**, if available, using reliable sources such as the Protein Data Bank (PDB).

From the **PDB** search interface, we can look for our target protein and refine the results by taxonomy, organism, method of structure determination, resolution, release date, and more.

For this example, we searched for PPAR-γ, applied a filter for Homo sapiens, and sorted the results from highest to lowest resolution. This yielded **479** structures, from which we will choose the most suitable entry based on biological relevance and structural quality.

> **Note:**
> - The PDB is a public repository that currently hosts over **240,000** three-dimensional structures of biomolecules from a wide range of organisms. - In cases where no experimentally resolved structures are available in the PDB (e.g., X-ray, cryo-EM, or NMR), alternative approaches such as homology modeling or computational prediction (e.g., AlphaFold) must be employed. These procedures, however, fall outside the scope of this tutorial.

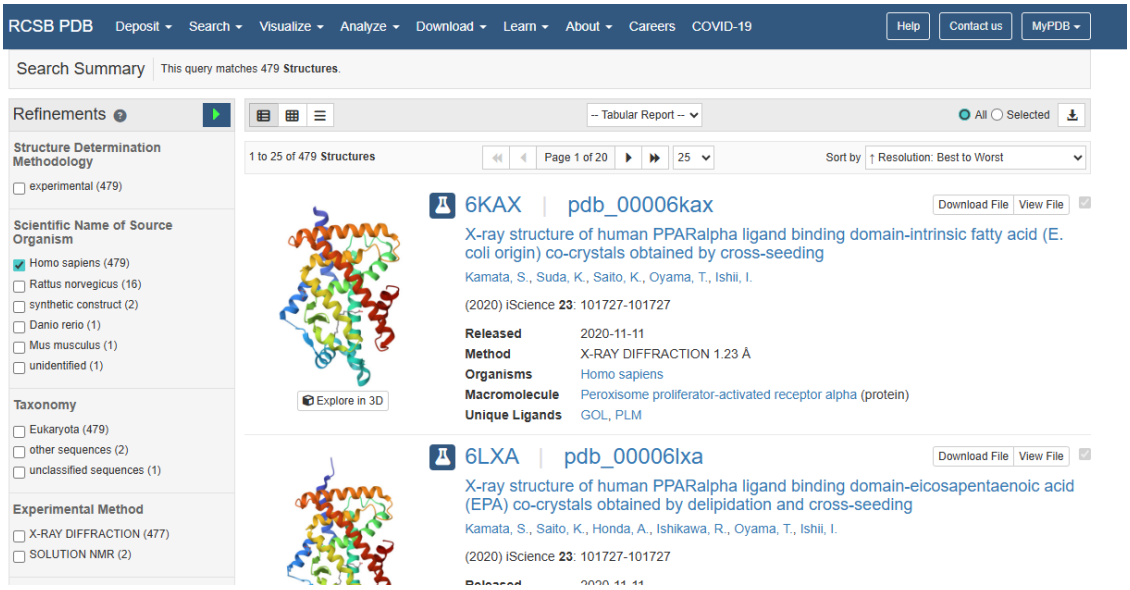

Figure 1: PPARG query in the PDB showing 479 Homo sapiens structures.

From here, we must review the results list and inspect candidate structures carefully. Be cautious: search queries often return **co-activators**, **DNA/RNA complexes**, or other assemblies that may not match the intended target.

Once a suitable entry is identified, open its record and examine the metadata. You should verify:

- **Authors** and **submission/release dates**
- **Structure quality** (method, resolution, refinement statistics)
- **Co-crystallized molecules** (ligands, cofactors, stabilizers, ions)
- **Primary citation** (original article)
- **Functional conformation** (e.g., active vs. inactive) and binding state
- **Sequence features** (mutations, deletions, missing loops)
- **Geometry/validation** (e.g., acceptable Ramachandran statistics)

For this tutorial, we will use a **PPAR-γ** structure in its **active conformation** with the PDB entry 5YCP.

On the entry page, select **`Download Files`** → **`Legacy PDB Format`**. Then, download the **`*.pdb`** structure from the PDB. Save the file to your working directory; in this tutorial, it is called **`5YCP.pdb`** (keep it as the raw reference).

Such a file contains the 3D coordinates of the protein, co-crystallized ligands (e.g., small organic molecules), co-activators (e.g., peptides), and other heteroatoms (e.g., ions, solvent).

> **Note:**
> - Although **PDBx/mmCIF** is the modern, preferred format, many downstream tools in docking pipelines still expect **Legacy PDB**. We use **`5YCP.pdb`** here for compatibility, and we will convert or clean it in later steps as needed.
> - We recommend consulting the original publication associated with the selected **PDB entry** to verify that the system is appropriate and to review the reported structural details, including the interactions with co-crystallized ligand(s).

Selection of the Ligand

Here we will work with a single ligand: **RGZ**. We need to obtain its 3D structure. There are several common molecular representations—**SMILES**, **InChI**, **SDF** (Structure Data File), **PDB**, **MOL2**, etc. You may use whichever format best fits your workflow; however, for molecular docking, starting from an `*.sdf` file is often more reliable and tends to introduce fewer errors.

A practical way to retrieve curated `*.sdf` files is through PubChem, which is a large public database containing information on **over 100 million** chemical substances, typically including structural data, physicochemical properties, vendor information, bioassay results, and more.

We can search **PubChem** using the common name of the compound **rosiglitazone** (i.e., **RGZ**) and then carefully select the corresponding entry. In this case, the entry for **RGZ** is **PubChem CID 77999**. Navigate to the **Structure** section and locate the adequate `3D Conformer` section → `DOWNLOAD COORDINATES`. Then, save the `*.sdf` file in your working directory. For this tutorial, it was called `rgz.sdf`.

Figure 2: 3D conformer download of Rosiglitazone from the PubChem server.

Note:
- In this example, we will manually search for the structure of rosiglitazone (i.e., `rgz.sdf`). For cases involving a large list of ligands, you can use the **PubChem API** to build a Python script that automatically fetches SDF files and associated metadata. - To verify correct atomic connectivity, the `rgz.sdf` file can be opened using a graphical viewer such as **PyMOL** or **UCSF Chimera** . - For highly flexible molecules, a **3D** conformer may not be available in PubChem; in that case, use the **2D** record instead.

2.2 Preparation of Structures

Now that we have our raw structures, we need to prepare them in the correct format and ensuring they have the proper structure for docking.

Preparing the Receptor Structure

In **PyMOL**, open the downloaded **`5YCP.pdb`** file from PDB. In some cases, you can notice the file must include an incomplete protein structure, the co-crystallized ligand, and other heteroatoms (e.g., ions, solvent). The goal is to produce one files with **only the receptor**. Furthermore, we will prepare another file with **the crystallized agonist**.

Select the relevant receptor chain (e.g., Chain **A**) and remove everything else. Save the result as **`5YCP_receptor.pdb`**.

```
PyMOL > select chain B
PyMOL > remove sele
PyMOL > select HOH
PyMOL > remove sele
```

Then isolate the ligand (**RGA**, **BRL**, etc) and save it as **`5YCP_ligand.pdb`**.

Figure 3: Raw structure as downloaded from the PDB 6MD4. In the figure below, the receptor is shown in blue, the ligand in green, and the co-crystallized components to be removed in red.

If the receptor structure contains **unresolved residues/atoms** that could not be refined in the crystallographic study, we will rebuild the receptor by **homology** using the **canonical sequence**. First, retrieve the canonical sequence from the UniProt database.

Search for **human PPAR-γ** and open entry **P37231-2**. In the **Sequences** section, copy the **canonical** sequence for downstream modeling.

CHIMERA is used to open the sequence and copy it to a **`*.ali`** file, as follows.

> **Note:**
> - UniProt is a public database that hosts **over 250 million protein records**.

Aligning the Canonical Sequence to the Crystal Template

If the crystal structure is incomplete (missing residues/atoms), we will rebuild the receptor by **comparative (homology) modeling** with **MODELLER** using the **canonical UniProt sequence** as the target and the **clean PDB** as the template.

Alignment file (`.ali`, PIR format).
To generate an alignment file, you may use the example provided in the `docking/` folder, **`seq_5YCP.ali`**, which contains two entries:

- The **template**, correponding to the crystallographic structure extracted from **`5YCP.pdb`** in Chimera or PyMOL, and

- The **target**, corresponding to the **canonical human PPAR-γ** sequence from UniProt (**P37231-2**).

Please, note that the **`*.ali`** files follow a highly specific PRI format, in which sequence lines are typically arranged in blocks of **81 characters**.

General conventions you should follow:

- The file uses **PIR-style headers** required by **MODELLER**, with entries formatted as `>P1;identifier`.
  - The **template** entry typically begins with `structureX:` and includes the **PDB code**, **start and end residue positions**, and **chain identifier**.
  - The **target** entry begins with `sequence:` and includes the identifier of the target sequence.
- **Gaps** corresponding to **missing residues** in the template should be represented by **-** characters in the alignment.
- If full-length modeling is unnecessary for the docking study, only the **crystallized region** of the protein should be aligned, for example, the **ligand-binding domain (LBD)** of **PPAR-γ**.
- Each sequence must end with an asterisk, `*`.

  Tip: Reading the original PDB article helps you decide which regions are functionally relevant. Crystal structures often lack flexible **loops/termini** (no defined secondary structure), so do not force-model those unless needed.

Once **`seq_5YCP.ali`** is ready, run your **MODELLER** script (e.g., **`make_model.py`**) pointing to:

- `alnfile` = `seq_5YCP.ali`
- `knowns` = template identifier (e.g., `5YCP_receptor`)
- `sequence` = target identifier (e.g., `PPARG_P37231-2`)
- `n_models` = number of models to generate (you will later choose the best by DOPE/other scores)

In **AutoModel**, `n_models` is set via the index range:

```
a.starting_model = 1
a.ending_model   = 3    # generates three models.
```

Run from your working directory:

```
python make_modelo.py
```

The script will generate one or more **PDB models** for the receptor's crystallized region. You can obtain typical outputs (filenames depend on sequence):

Filename	molpdf	DOPE score	GA341 score
PPARG_P37231-2.B99990001.pdb	1173.57715	-35933.17969	1.00000
PPARG_P37231-2.B99990002.pdb	1117.48389	-36061.49219	1.00000
PPARG_P37231-2.B99990003.pdb	1189.37024	-35883.63281	1.00000

This step outputs the specified number of models and their scores. Select the **top-scoring** one (e.g., by **DOPE**).

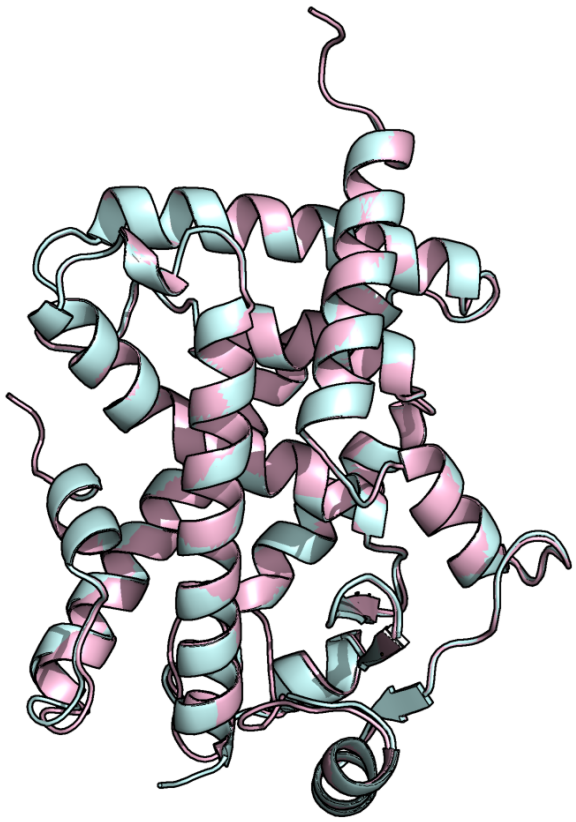

Figure 4: Alignment between the raw structure and our modeled protein.

You can also open the models in **PyMOL** and align them to the cleaned crystal structure to inspect **RMSD** and visualize differences in the **modeled regions**. In our case, we compared the modeled protein (**pink**) with the cleaned **5YCP** structure (**blue**) and obtained an **RMSD of 0.090 Å**, indicating excellent agreement. The main deviations occur in **loop regions**; notably, the previously missing loop was modeled toward the

orthosteric site, which could interfere with docking. Keep this in mind when selecting structures. Once satisfied, choose the model for the next steps and proceed with receptor preparation for docking.

The numbering for amino acid should be homogolized with **UNIPROT** using **pdb-tools** as follows:

```
pdb_reres -204 PPARG_model.pdb > PPARG_model_reres204.pdb
```

Binding-Site Identification

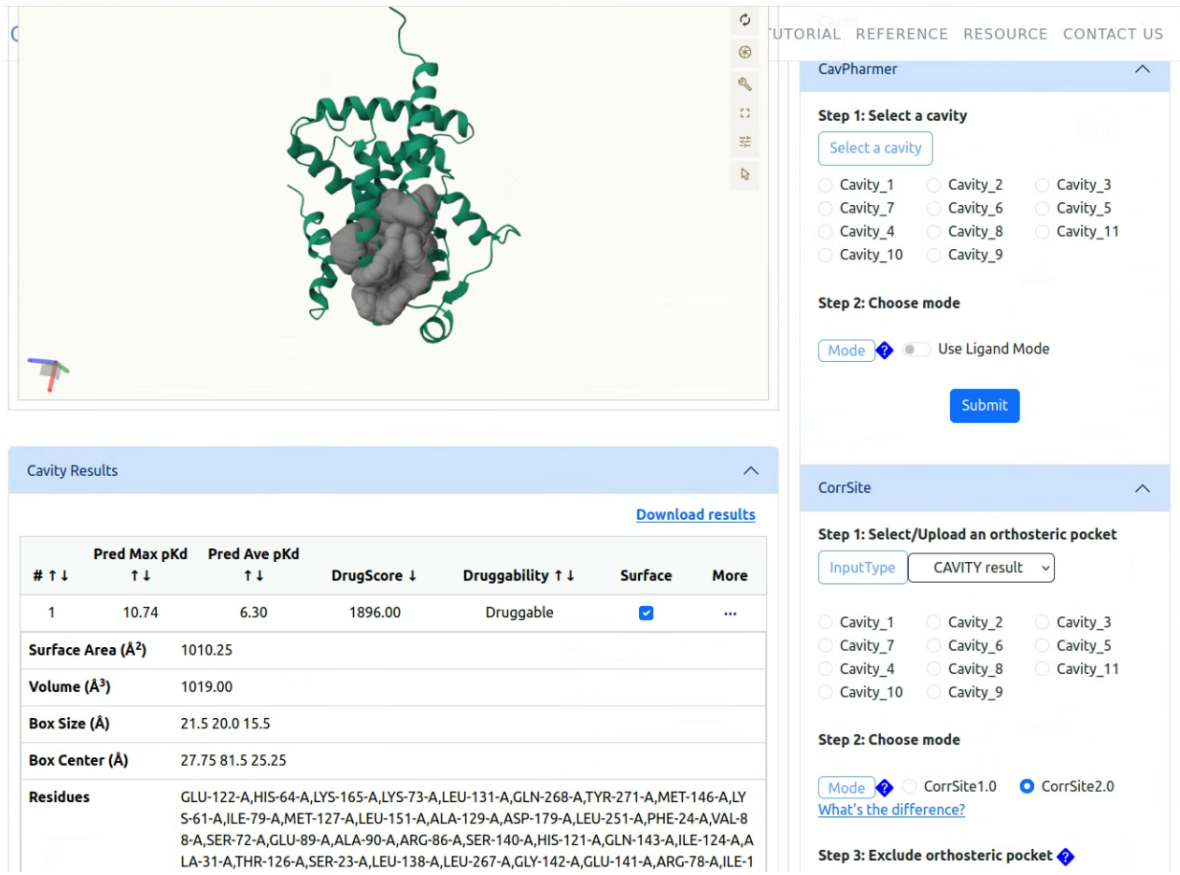

Figure 5: Main cavity calculated with Cavity Plus.

Once the receptor model is finalized, submit the **PDB** file to Cavity Plus server to identify candidate binding pockets. The server ranks cavities by **druggability score** and reports **volume**, **surface area**, **centroid coordinates**, and **lining residues**.
These details help (a) pick the pocket to target and (b) define the docking box center.

What to record from CavityPlus

- Pocket ID (rank)
- Centroid coordinates (x, y, z)
- Key residues (for later interaction checks)
- Pocket volume/surface (optional context)

Tip: Prefer the top-ranked pocket that matches the known orthosteric site (if applicable).

Docking Search Space (Grid Box)

Define the Vina/Qvina search space based on the selected pocket:

- **Center**: use the CavityPlus pocket centroid

```
--center_x <x> --center_y <y> --center_z <z>
```

- **Size (Å)**: set box edges to encompass the pocket + ~4–6 Å margin *(example values)*

```
--size_x 21 --size_y 20 --size_z 16
```

Example (placeholder numbers)

```
--center_x 10.2 --center_y 14.7 --center_z 7.9
--size_x 21 --size_y 20 --size_z 16
```

Note:
- Keep the box as tight as practical around the binding site to reduce false positives. - For multiple receptor states, keep **center/size** consistent unless pocket geometry differs significantly.
- Document the final **center** and **size** you use; you will need them for reproducibility and for plotting/analysis later.

Preparing the Ligand Structure

To prepare the ligand, the **`*.sdf`** file must first be converted to **`*.pdb`**, followed by an energy minimization step, and then converted to **`*.pdbqt`** for docking. This procedure requires the use of Babel, which is installed in a separate environment named **`*py2`**. Accordingly, before running Babel, this environment must be activated from the terminal with the following command:

```
conda activate py2
```

Now, we can use Babel to convert our SDF file to PDB, ensuring that the structure has hydrogens:

```
babel rgz.sdf -opdb -O rgz.pdb -h
```

We should visually inspect the output .pdb file to see that the connectivity and hydrogens are correct. Once we are sure that everything looks good now we will use obminimize to perform an energetic minimization of the structure using a force field. There are 5 possible force fields: GAFF, MMFF94, MMFF94S, Ghemical, and UFF, but I have found best results using MMFF94 (or GAFF). The number of steps of minimization also needs to be specified, 5,000 could be a good start for tests, but I like to do 20,000 steps to ensure enough steps to achieve a minimum, specially for flexible molecules. To do the minimization write in the terminal:

```
obminimize -n 20000 -ff MMFF94 rgz.pdb > rgz_min.pdb
```

Again, we should visually inspect the output to ensure the structure has the correct connectivity and valences. Finally, we need to convert our minimized .pdb structure into the expected .pdbqt file for docking with Autodock Vina. We will write in the terminal the following command:

```
babel rgz_min.pdb -opdbqt -O rgz.pdbqt
```

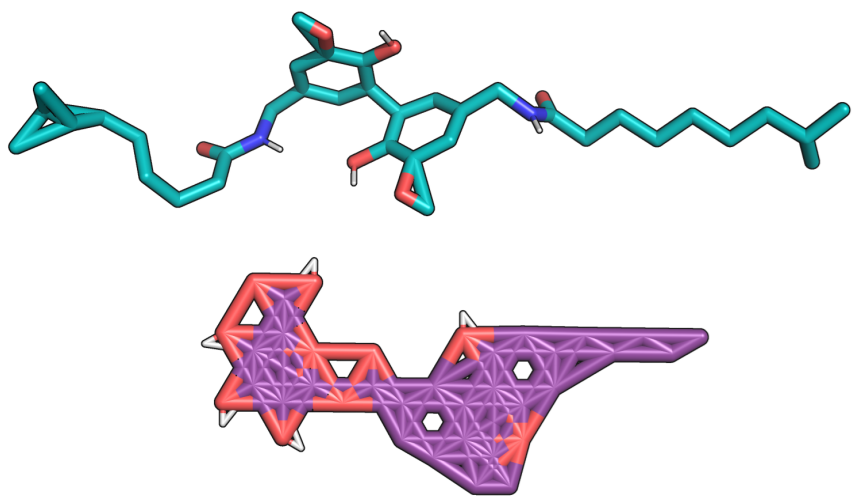

Figure 6: Bad connectivity in the converted .pdbqt file.

Before moving on we should (again) visually inspect the structure. In the .pdbqt files the non-polar hydrogens are not explicit, thus we will only see the hydrogens connected to non-carbon atoms. It is important to visually inspect the files, as sometimes we can see connectivity issues such as the example in Figure 5 (down). In this case, the structure of the ligand is not correct, and it will not be able to dock properly. This could be due to a problem with the force field used for minimization, or it could be due to a problem with the conversion to .pdbqt. When seeing this, we should try to change the force field on the minimization step. If that does not work then we can add the -gen3d argument to the obminimize line

```
obminimize -n 20000 -gen3d -ff MMFF94 rgz.pdb > rgz_min.pdb
```

2.3 Docking Setup and Execution

Now we will need to finish our setup before performing the docking. Here we need to get the structure of the receptor in .pdbqt, as well as defining a docking box and other docking parameters. We will prepare the .pdbqt of the receptor and the docking box in one step using Chimera. We will use the modeled PPAR γ (PPARG_model.pdb) and the rosiglitazone from the crystal (6MD4_ligand.pdb). Both structures will be opened in Chimera, then we will go to the "Tools" tab, then to the "Surface/Binding Analysis" tab, and finally select "AutoDock Vina". This will open a dialog box where we will need to define the output location and name (PPARG.pdbqt), select the structure corresponding to the receptor, and the one corresponding to the ligand.

Then, in the center and size boxes is where the dimensions of the box will be defined, in cases where we have no idea where to begin we can start at center 0,0,0 and size 20,20,20. Then we will need to resize and

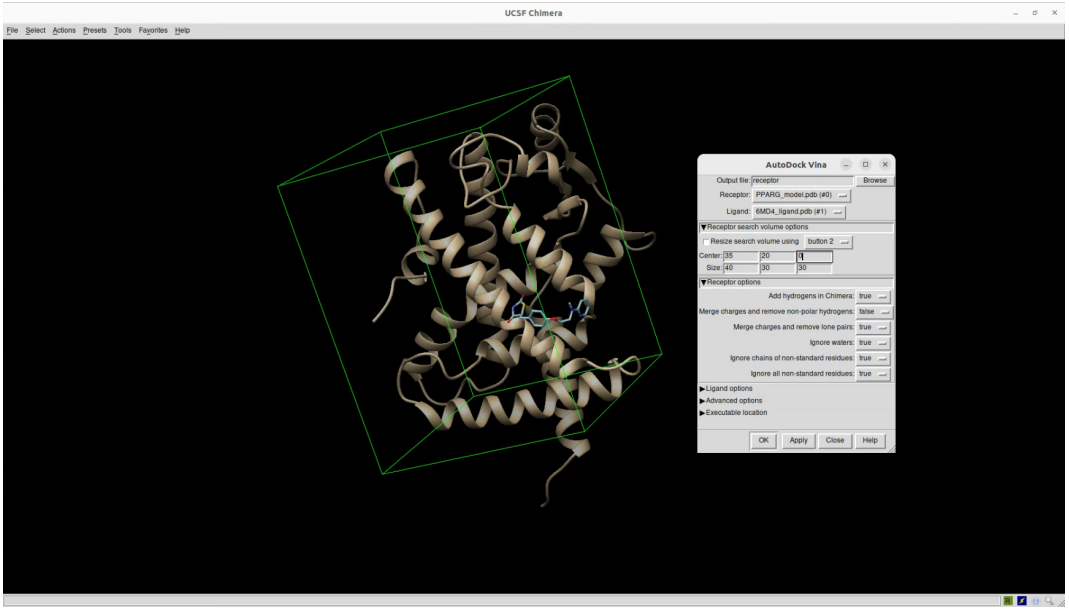

Figure 7: Definition of the docking configuration in Chimera.

define the box according to our needs; if we are interested in interactions with a particular region of the receptor (e.g. orthosteric site) then the box should encompass that region, in cases where we do not know where the ligands can interact then we can build a box that encompasses the whole receptor.

> **Note:**
> Do not make the box too tight, leave some room for error as some ligands could sneakily fit when we are restraining.

After defining the right box, we will click OK. Here, Chimera will attempt to dock the selected ligand to the receptor, and preparing a .pdbqt file of both the ligand and receptor, and also adds hydrogens. We will also see a .conf file that was generated, this will become the configuration for our docking, here we see our box dimensions and three more parameters. Exhaustiveness means how extensively the docking will be, the default if 8, but I like to use 100 for more precise calculations. Energy range is the maximum energy difference between the best pose and the rest of reported poses. Finally num of poses is how many poses will report, the default is 10, but they could be reduced.

So, now we have our two **`*.pdbqt`** structures for the receptor and ligand, and our **`*.conf`** file, and we are ready to perform our first docking.

Perform the Docking

If everything else went well, then this should be the easiest part. To perform the docking we will use the following command:

```
../../qvina2.1 --receptor PPARG.pdbqt --ligand rgz.pdbqt --config pparg_dock.conf
--out rgz_docked.pdbqt --num_modes 1 --log rzg.log
```

And *voilà*, there we have our docking. We can see on the terminal the docking binding energy in kcal/mol.

```
##############################################################################
# If you used Quick Vina 2 in your work, please cite:                        #
#                                                                            #
# Amr Alhossary, Stephanus Daniel Handoko, Yuguang Mu, and Chee-Keong Kwoh,#
# Fast, Accurate, and Reliable Molecular Docking with QuickVina 2,           #
# Bioinformatics (2015), doi: 10.1093/bioinformatics/btv082                 #
#                                                                            #
# You are also encouraged to cite Quick Vina 1:                              #
# Stephanus Daniel Handoko, Xuchang Ouyang, Chinh Tran To Su, Chee Keong     #
# Kwoh, Yew Soon Ong,                                                        #
# QuickVina: Accelerating AutoDock Vina Using Gradient-Based Heuristics for#
# Global Optimization,                                                       #
# IEEE/ACM Transactions on Computational Biology and Bioinformatics,vol.9, #
# no. 5, pp. 1266-1272, Sept.-Oct. 2012, doi: 10.1109/TCBB.2012.82           #
#                                                                            #
# and original AutoDock Vina paper:                                          #
# O. Trott, A. J. Olson,                                                     #
# AutoDock Vina: improving the speed and accuracy of docking with a          #
# new scoring function, efficient optimization and multithreading,           #
# Journal of Computational Chemistry 31 (2010) 455-461                       #
# doi: 10.1002/jcc.21334                                                     #
##############################################################################

WARNING: The search space volume > 27000 Angstrom^3 (See FAQ)
Detected 24 CPUs
Reading input ... done.
Setting up the scoring function ... done.
Analyzing the binding site ... done.
Using random seed: 1034255104
Performing search ... done.
Refining results ...
searching finished in 16.140 seconds
done.

mode |   affinity | dist from best mode
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
-----+------------+----------+----------
   1         -8.6      0.000      0.000
   2         -8.5      4.239      9.638
   3         -8.3      3.282      4.848
Writing output ... done.
```

Figure 8: The terminal should show this docking log after running the docking command.

2.4 Analysis of Results

We can open in Pymol the used structure and the output _docked.pdbqt file to see its best docking pose. We can also create some pretty nice figures with Pymol to showcase where the preffered pose of the ligand was on blind docking. To get the egenral figure of the blind docking we can open our receptor's **`*.pdbqt`** file and all the _docked.pdbqt ligands in Pymol, we can also include the surface of the calculated cavity to showcase if the best docking poses fit within this orthosteric site. In the following figure we see the receptor in blue, the docked molecules in orange and the calculated binding site in yellow.

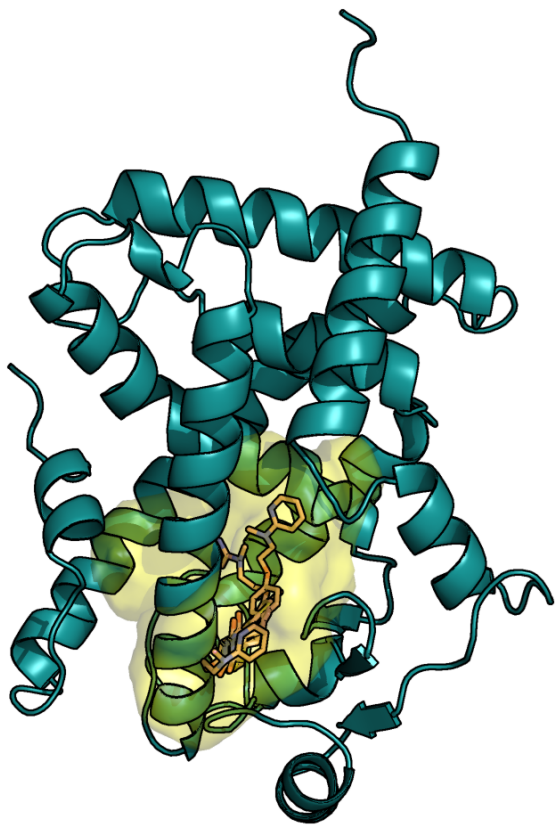

Figure 9: Best poses after docking.

To get the aestethic of this figure, I first changed the background to white, then changed the ray trace mode with this command:

```
set ray_trace_mode, 1
```

then I set the ray trace gain to show the black outline with:

```
set ray_trace_gain, 0.005
```

And I changed the presentation of the cavity to surface and changed its transparency to 0.5 with:

```
set transparency, 0.5
```

If we want a more thorough analysis of the interactions we can save a **`*.pdb`** file with the receptor and the docked molecule, then open it in LigPlot+ to see the relevant residues and types of interactions. We will see the residues that interact with the molecule, with those red outlines indicating hydrophobic interactions, and the H-bonds will be indicated with green dashed lines.

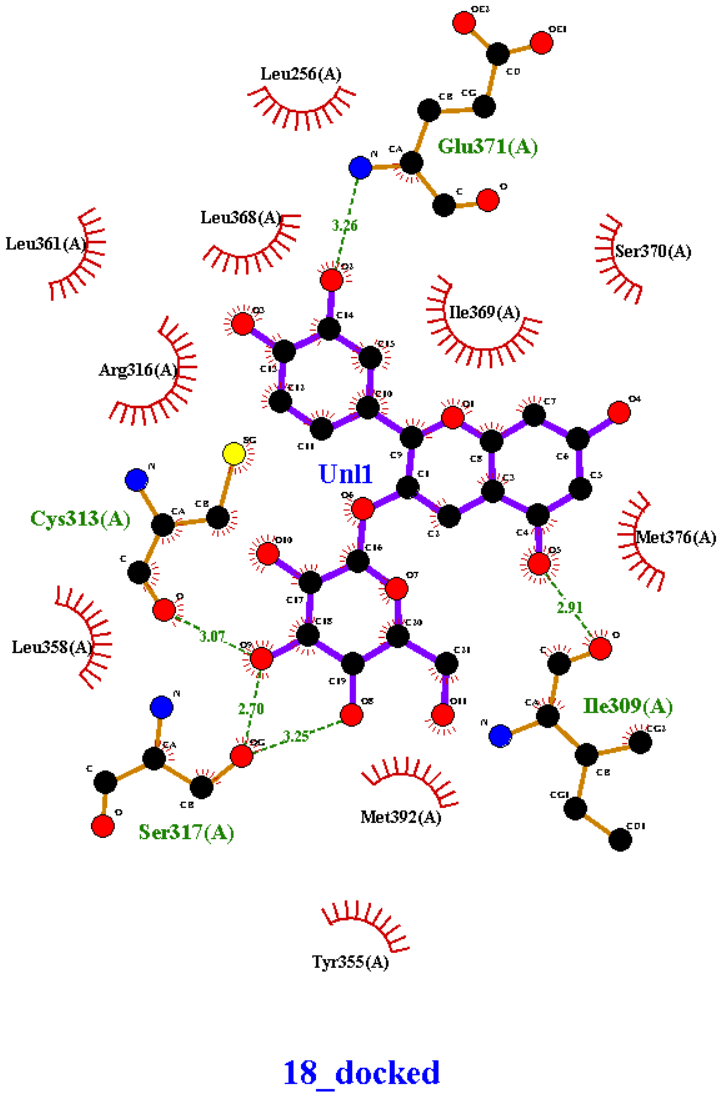

Figure 10: LigPlot+ shows the interactions of the docked molecule.

Note:
Be careful when dealing with the numbering of the receptor, we should usually use the canonical numbering, but when making new models the numbering can change and it will affect when we report our results and the numbers do not match the ones reported in the reference article

from the crystal structure.

Now that we now the relevant interactions, we can make figures of the closeups of the molecule, showing the interacting residues. To achieve this, we can first change the transparency of the cartoon to avoid confusion when showcasing the molecule.

```
set cartoon_transparency, 0.7
```

Then we can show the distances between the ligand and the protein within a cutoff distance in Å. You can choose any value, but the larger the distance the more interactions there will be and the figure can look saturated. In this command objects 1 and 2 should correspond to the name of the structures that we want to showcase.

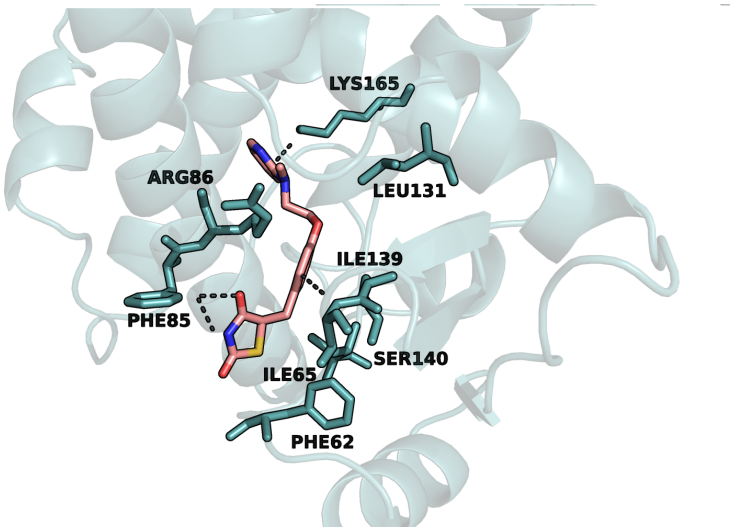

Figure 11: Closeup of the docked molecule showcasing its interactions with the relevant residues within 2.5 Å.

```
distances polar, object_1, object_2, 2.5
```

We can change the color of the interaction lines, and also show or hide the distance labels. Next we will need to select the interacting residues within that same cutoff distance with tha following command, where "LIG" is the name of our docked molecule:

```
select br. all within 2.5 of LIG
```

Now we will have selected the relevant residues. Finally, we can show in licorice that selection, hide hydrogens and valences for clarity purposes, and add labels:

```
label n. CA and i. 235, "%s%s" % (resn,resi)
```

This will label the selected interacting residues in their alpha carbons, and include the residue name and number. Now we can find the best position for our final figure, and click on the left bottom corner in the pymol window and click where it says "3-button viewing", this should change the mode to "3-button

editing". Now we can move the labels by doing ctrl+right click and positions the labels in a way that they are readable. Finally, we adjust our ray trace settings and export our figure that should look something like Figure 11.

We can also do repetitions of the same experiment, or get more docking poses to get a statistical analysis of the docking energies and poses. We can do the docking on other conformations of PPAR γ, and analyze the differences between the energies and interactions with those conformations. We can also weight these energies on another parameter to get the Ligand efficacy metric to make more practical sense of the results.

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